KEY MACROECONOMIC VARIABLES CHAPTER 2. BLANCHARD

MACROECONOMICS

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Reinventing Higher Education


IN WHICH COUNTRY WOULD YOU LIKE TO LIVE? WHY?
ARCTIC OCEAN
United States
Canada
United States
Mexico
Caribbean
Central America
Guyana
Suriname
French Guiana (France)
Venez.
Colombia
Ecuador
Peru
Brazil
Bolivia
Paraguay
Chile
Argentina
Uruguay
PACIFIC OCEAN
ATLANTIC OCEAN
Iceland
Norway
Sweden
Finland
Russia
U.K.
Ireland
Den.
Neth.
Belg.
Germany
France
Spain
Portugal
Poland
Belar.
Ukraine
Mol.
Rom.
Bul.
Hung.
Cz.
Estonia
Latvia
Lithuania
Italy
Greece
Turkey
Cyprus
Armenia
Georg.
Az.
Kazakhstan
Uzbek.
Turkm.
Iran
Iraq
Syr.
Lebanon
Israel
Jordan
Kuwait
Qatar
Saudi Arabia
U.A.E.
Oman
Yemen
Afghan.
Pakistan
Mongolia
Kyrgyzstan
Tajikistan
China
India
Nepal
Bhutan
Bangl.
Laos
Vietnam
Cambodia
Thai.
Bur.
Taiwan
Philippines
N.Korea
S.Korea
Japan
Sri Lanka
Malaysia
Indonesia
Papua New Guinea
Timor-Leste
Solomon Islands
Fiji
New Caledonia
Australia
New Zealand
Tunisia
Morocco
Western Sahara
Algeria
Libya
Egypt
Maurit.
Mali
Niger
Chad
Eritrea
Sudan
Djibouti
Senegal
The Gambia
Guinea-Bissau
Guinea
Sierra Leone
Liberia
Cote d'Ivoire
B.Faso
Ghana
Togo
Benin
Nigeria
Cam.
C.A.R.
South Sudan
Ethiopia
Somalia
Ken.
Dem. Rep. of Congo
Congo
Equatorial Guinea
Gabon
Uganda
Rwanda
Burundi
Tanz.
Angola
Zambia
Zim.
Bots.
Namibia
South Africa
Lesotho
eSwatini
Malawi
Mozambique
Madagascar
INDIAN OCEAN



IF YOU WERE AMONG THE POOREST 10%, IN WHICH COUNTRY WOULD YOU LIKE TO LIVE? WHY?

Outline/Learning Objectives

1. Aggregate Output = GDP
2. The Unemployment Rate
3. The Inflation Rate
4. Okun's Law: Output and Unemployment
5. Phillips Curve: Unemployment and the Inflation Rate
6. The Short Run, the Medium Run, and the Long Run

1. Aggregate Output= GDP

GDP

The market value of all final goods and services produced within the borders of a country during a particular time period.

GDP = Total Production = Total Expenditure = Total Income

Which is the GDP and GDP per capita from your country?

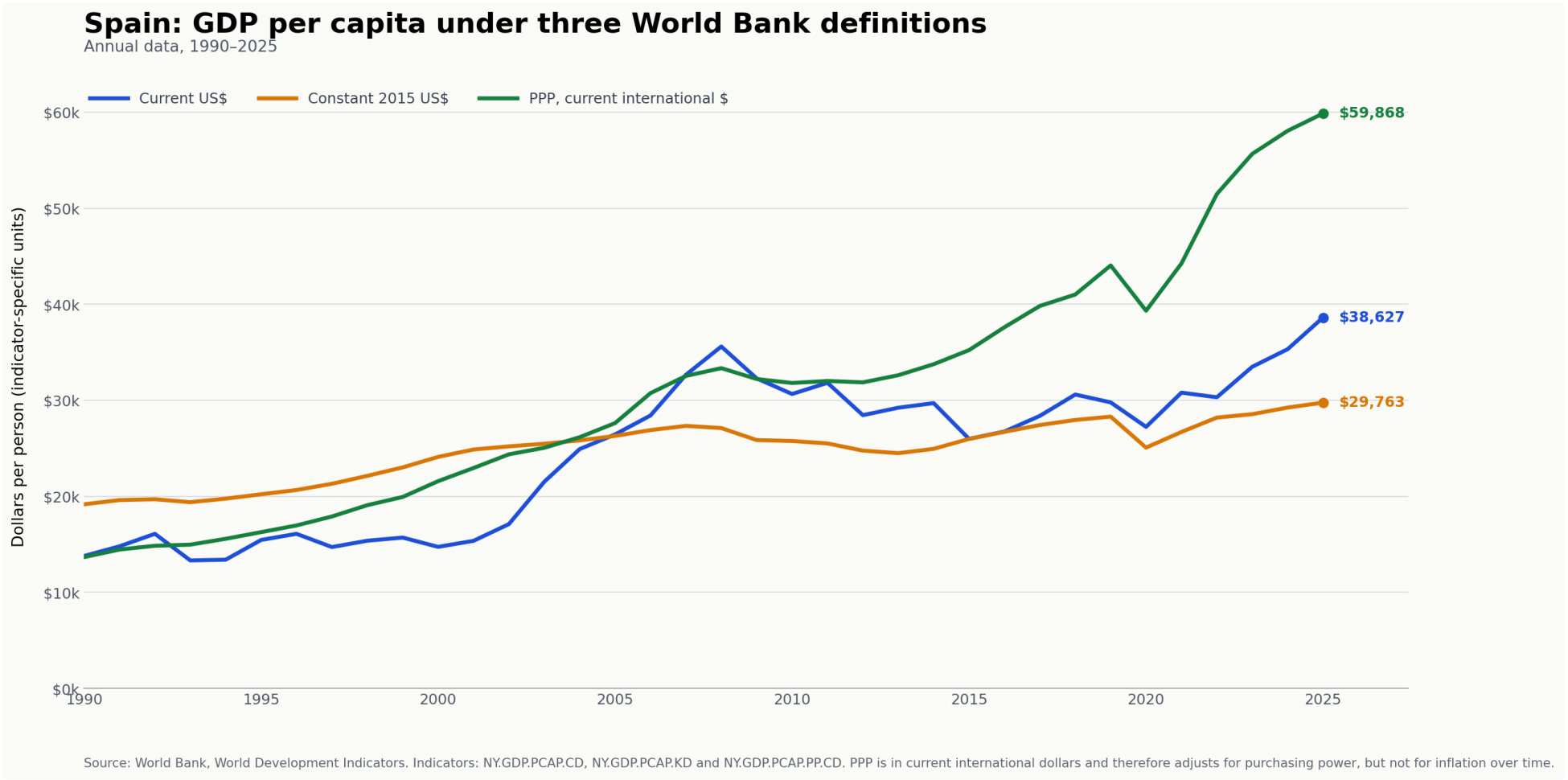
What are SPAIN´s GDP and GDP per capita?

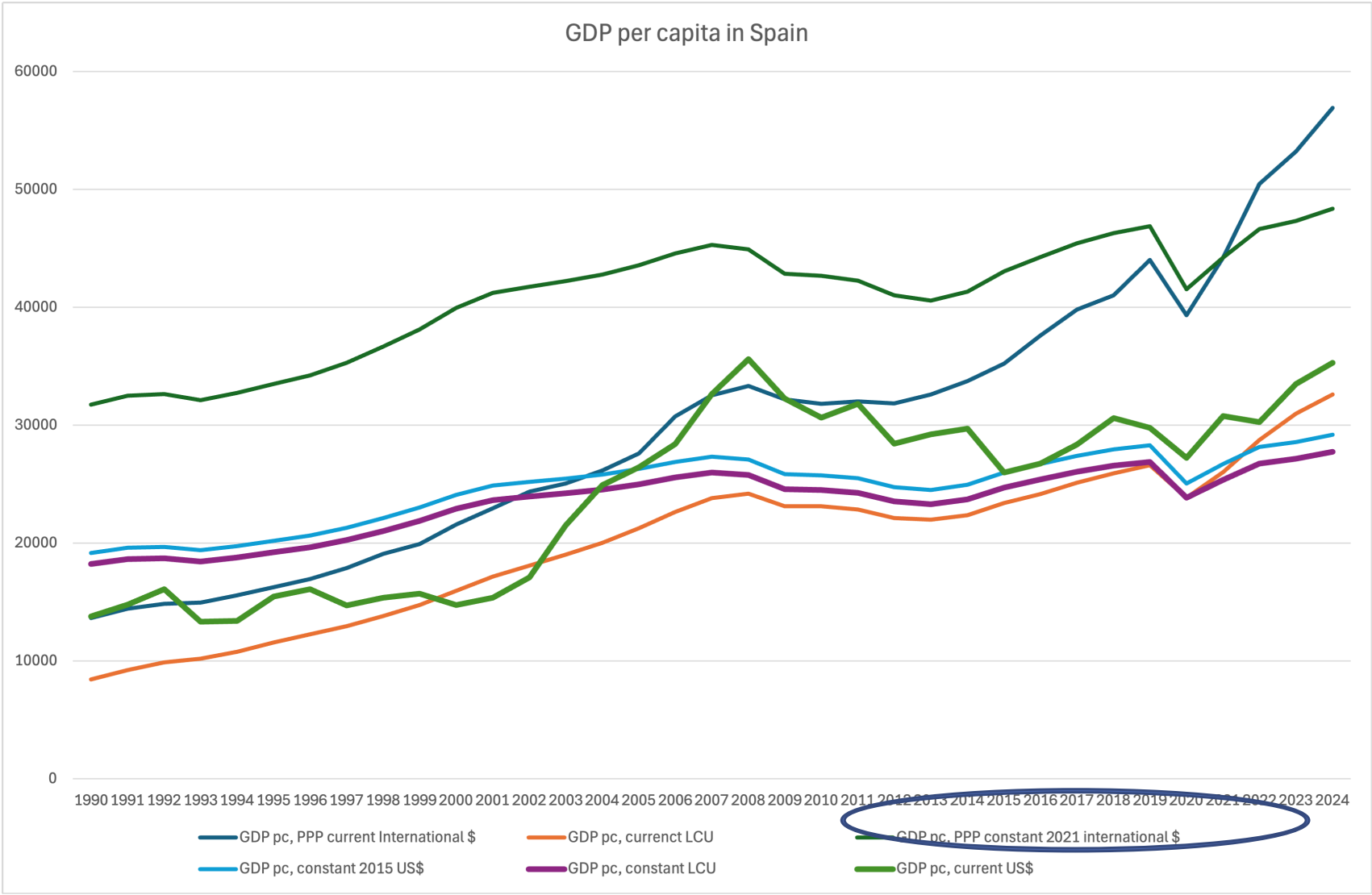
THE WORLD BANK
IBRD • IDA | WORLD BANK GROUP

https://data.worldbank.org

WHICH ONE TO CHOOSE?

GDP per capita, PPP (constant 2011 international $)
GDP per capita, PPP (current international $)
GDP per capita (constant LCU)
GDP per capita growth (annual %)
GDP per capita (constant 2010 US$)
GDP per capita (current LCU)
GDP per capita (current US$)
GDP, PPP (constant 2011 international $)
GDP, PPP (current international $)
GDP (constant LCU)
GDP growth (annual %)
GDP (constant 2010 US$)
GDP: linked series (current LCU)
GDP (current LCU)
GDP (current US$)


Spain: GDP per capita under three World Bank definitions
Annual data, 1990–2025
Current US$
Constant 2015 US$
PPP, current international $
Dollars per person (indicator-specific units)
$60k
$50k
$40k
$30k
$20k
$10k
$0k
1990
1995
2000
2005
2010
2015
2020
2025
$59,868
$38,627
$29,763
Source: World Bank, World Development Indicators. Indicators: NY.GDP.PCAP.CD, NY.GDP.PCAP.KD and NY.GDP.PCAP.PP.CD. PPP is in current international dollars and therefore adjusts for purchasing power, but not for inflation over time.


GDP per capita in Spain
60000
50000
40000
30000
20000
10000
0
1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024
GDP pc, PPP current International $
GDP pc, currenct LCU
GDP pc, PPP constant 2021 international $
GDP pc, constant 2015 US$
GDP pc, constant LCU
GDP pc, current US$

Real GDP (constant prices)

Nominal GDP adjusted for inflation

GDP per cápita (p/c)

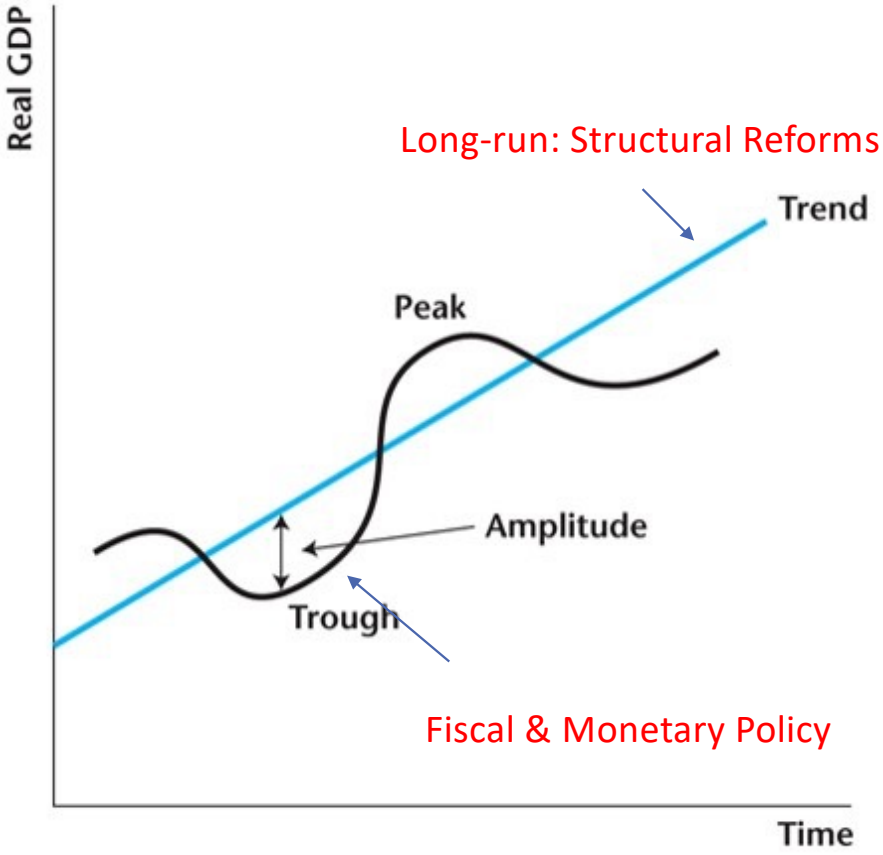
Value of total GDP divided by the number of people in the country.



For International Comparisons we use

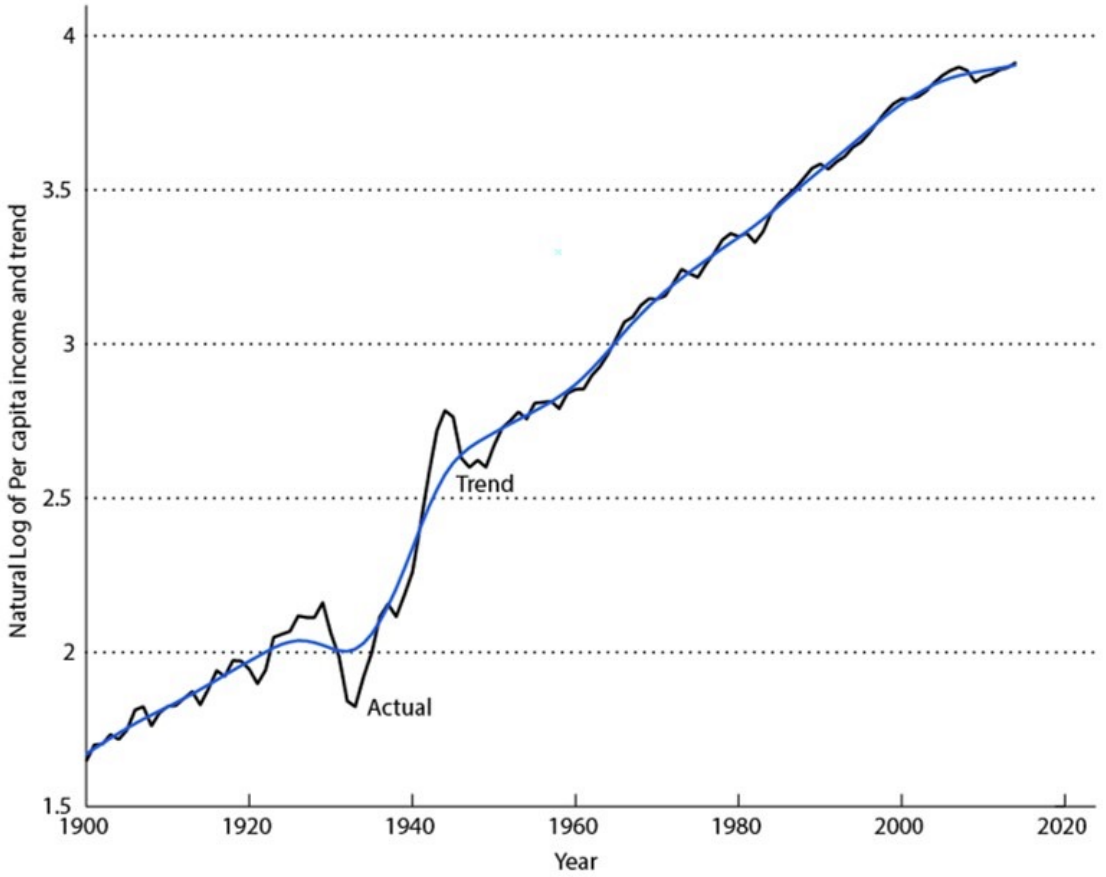
GDP adjusted for relative CPI (Consumer Price Index)= PPP (Purchasing Power Parity)

https://www.gapminder.org/tools/#$chart-type=linechart

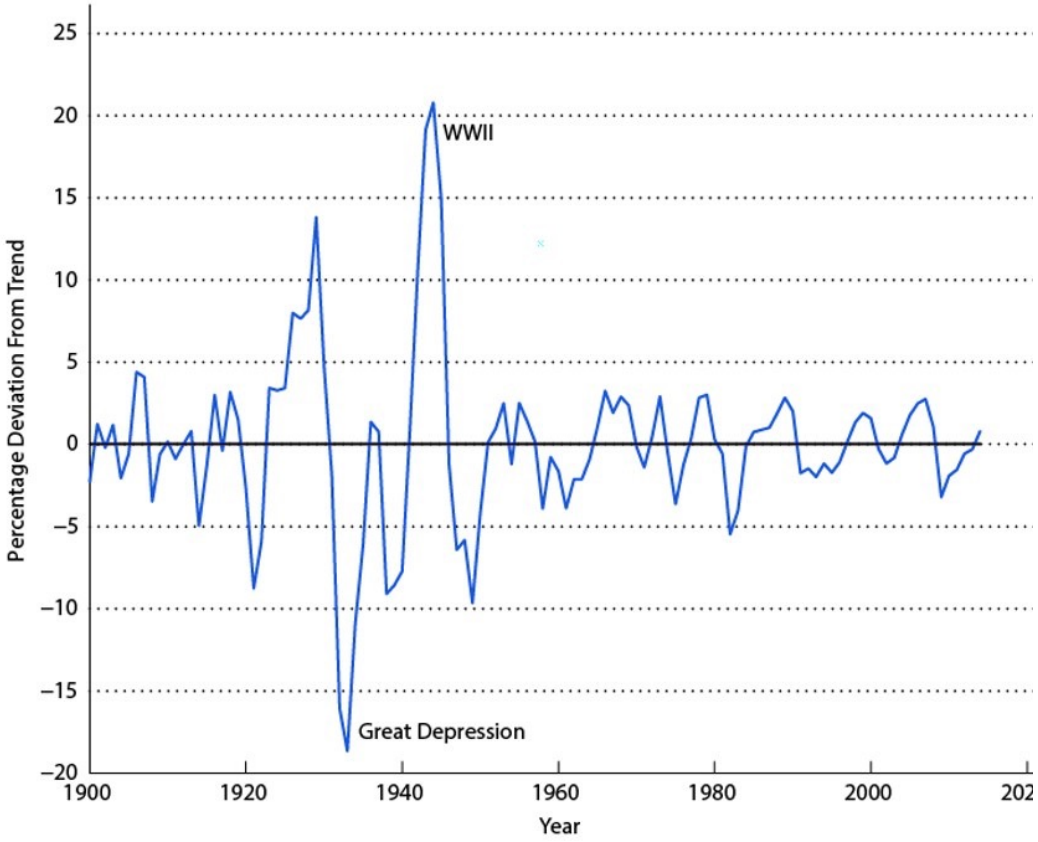


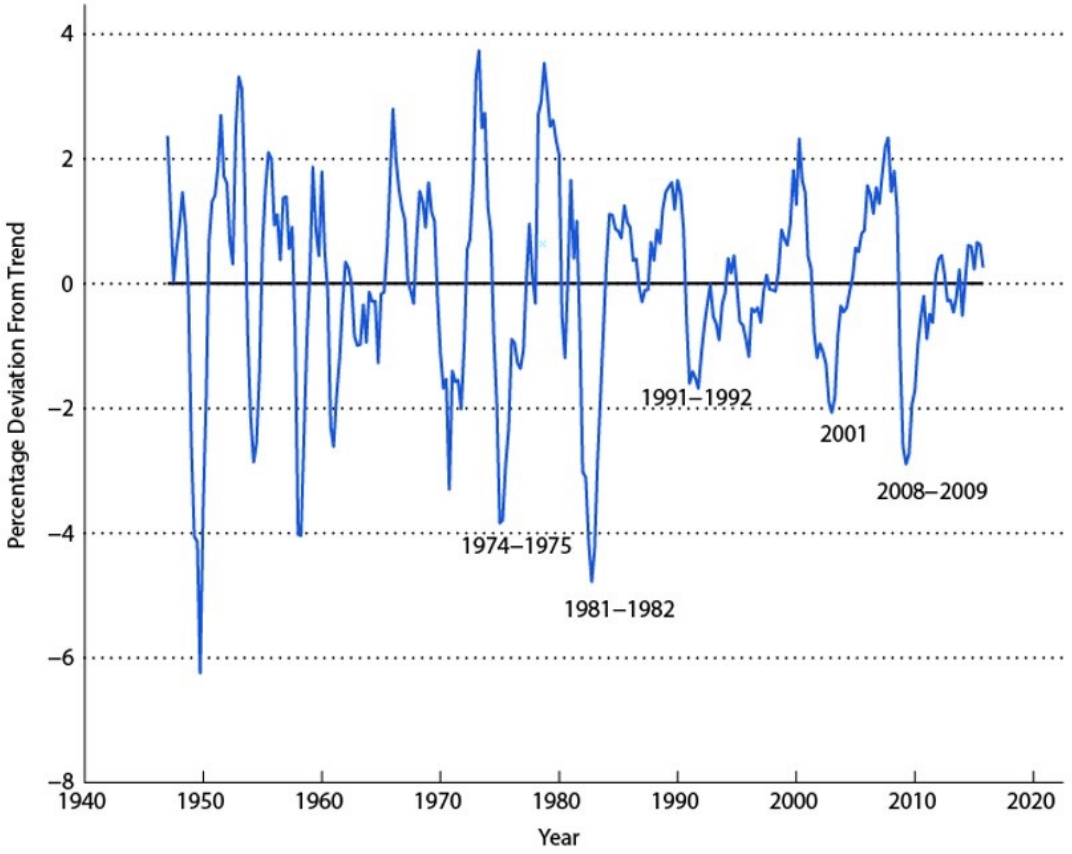
Source: Williamson (2018)

Natural Logarithm of Per Capita Real GDP and Trend



Percentage Deviations from Trend in Per Capita Real GDP



















What is INCLUDED in GDP

What is NOT included in GDP













Prostitution is included in GDP only where it is legal:
It is estimated in the EU, not in the US.



Prostitution and drugs account for €9bn of Spain's GDP, says statistics institute

Inclusion of illicit activities and other changes to calculation methodology boost 2013 total by 2.6%

ALEJANDRO BOLAÑOS

Madrid - 25 SEP 2014 - 12:42 CEST

https://elpais.com/elpais/2014/09/25/inenglish/1411640056_131333.html

- Illicit activities – prostitution, drug trafficking and smuggled goods – account for 0.87% of GDP.
- To estimate the value that drug trafficking, INE has taken into account the **quantity of narcotics seized by the police**, using that amount to estimate the total impact the production and sale of illegal substances has on the economy. A similar methodology has been used for smuggling.
- As for prostitution, the calculation is based on the number of people working in the country's **brothels**, which under Spanish law reside in a gray area, being neither fully legal nor illegal.



Exhibit 4 Underground Economy as a Percentage of Nominal GDP (2006)

Country	Underground Economy as a Percentage of Nominal GDP (%)
Peru	60.0
Mexico	32.1
South Korea	27.5
Costa Rica	26.8
Greece	26.0
India	24.4
Italy	23.1
Spain	20.2
Sweden	16.3
Germany	15.4
Canada	14.1
China	14.0
France	13.2
Japan	8.9
United States	8.0

Source: Friedrich Schneider and Andreas Buehm, Linz University, 2009.

Table 1 Summary statistics EVADE (share of GDP)

Country	Mean	Std. deviation	Country	Mean	Std. deviation
Greece	.36	.04	Slovakia	.12	.05
Italy	.31	.01	Hungary	.11	.04
Spain	.24	.07	Finland	.10	.02
Portugal	.24	.04	Czechia	.10	.06
Lithuania	.20	.03	Austria	.09	.01
Latvia	.20	.07	Denmark	.09	.01
Bulgaria	.20	.06	Estonia	.08	.04
France	.14	.02	Slovenia	.07	.02
Germany	.13	.01	Sweden	.06	.03
Croatia	.12	.03	Belgium	.05	.01

***Notes*: This table reports the average level of EVADE as defined in equation (4), and the within-country standard deviation of EVADE over the period 1999-2020.**
***Source*: Pappadà and Rogoff (2025).**

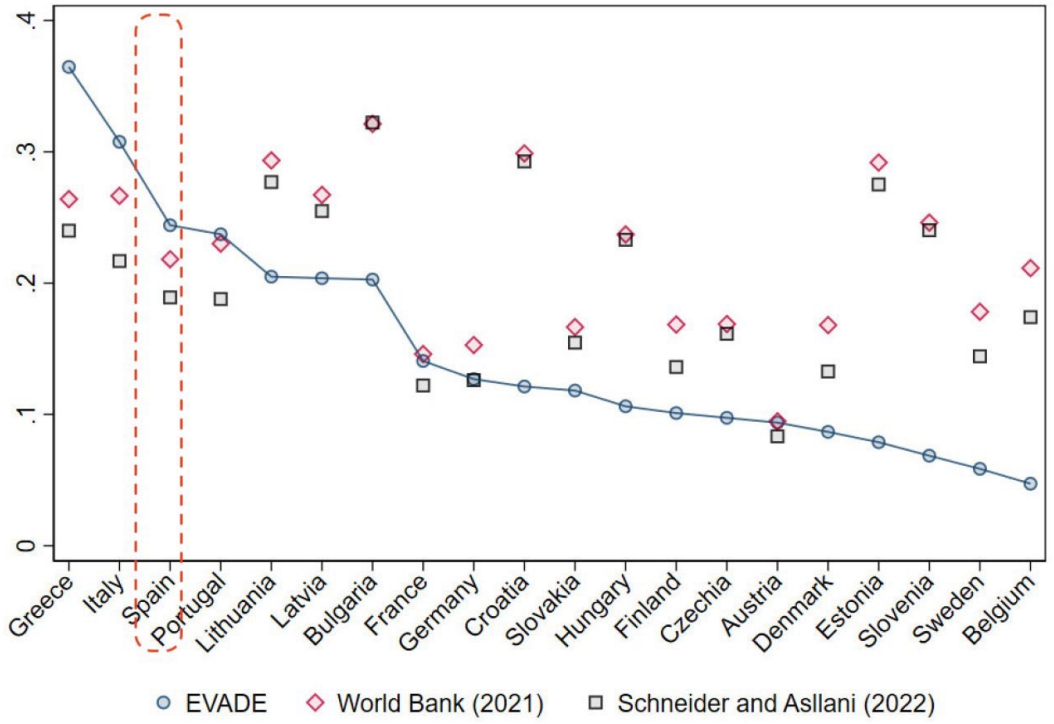


FIGURE 4. Alternative measures of informal economy—average level (share of GDP). This figure displays the average level of *EVADE* and alternative measures of informal economy such as the World Bank (Elgin et al. 2021), and Schneider and Asllani (2022) (MIMIC).

How much contributes a Professor to GDP during an hour of class?

- His wage.
- Therefore, if IE increases my salary 5%, “my GDP” increases 5%, even though I teach exactly the same.
- Imagine the Professor writes an article in a blog.
  It takes him an hour.
  How do we account for it? The opportunity cost?
  The same as one hour of teaching?
  What is the “wellbeing” that he creates?



ECON PRACTICE



What's included in U.S. GDP?

PLAY

https://practice.mru.org/gdp/?utm_source=EX&utm_medium=Graph&utm_campaign=MRUEmail&utm_content=GDP-InteractiveProf&__s=r7kmrfy33m1u6tdnfzv3

Transaction	GDP treatment	Why?
A new domestically made car	COUNTED	A current final good produced inside the country.
A used car is resold	PARTLY	The car is not recounted; the dealer’s service or margin is.
A newly constructed home	COUNTED	New residential construction is investment.
An existing home is resold	PARTLY	The house is not recounted; agent and legal services are.
A pension benefit is paid	NOT DIRECTLY	It is a transfer; goods and services later purchased may count.
A household buys an imported phone	ZERO NET	C rises, but M subtracts the same foreign-produced value.
An investor buys shares	PARTLY	The asset trade is excluded; brokerage services are included.
An owner lives in their own home	IMPUTED	National accounts estimate the housing service as imputed rent.

1. Aggregate Output

It can be measured in three different ways:

1. Production Approach
Production-based accounting sums up each firm's **value added**, which is the firm's sales revenue minus the firm's purchases of intermediate products from other firms.
2. Income Approach
Income-based accounting sums up payments (or income) received by labor and the owners of physical or financial capital.
3. Expenditure Approach
Expenditure-based accounting sums up the purchases of goods and services by different groups or categories.
There are five main categories: C+I+G+X-M

1. Aggregate Output

EXAMPLE 2

Consider an economy with two firms, Firm 1 and Firm 2.

Is aggregate output the sum of the values of all goods produced, i.e., $300? Or just the value of cars, i.e., $200?

Steel is an **intermediate good**, which is a good used in the production of another good.

Steel Company (Firm 1)		
Revenues from sales		$100
Expenses		$80
Wages	$80	
Profit		$20

Car Company (Firm 2)		
Revenues from sales		$200
Expenses		$170
Wages	$70	
Steel purchases	$100	
Profit		$30

1. Aggregate Output

- **Nominal GDP** is the sum of the quantities of final goods produced times their *current price*. Is also called **dollar GDP**, or **GDP in current dollars**.

  Nominal GDP increases for two reasons:
  - The production of most goods increases
  - The price of most goods increases

  Our goal is to measure production and its change over time.

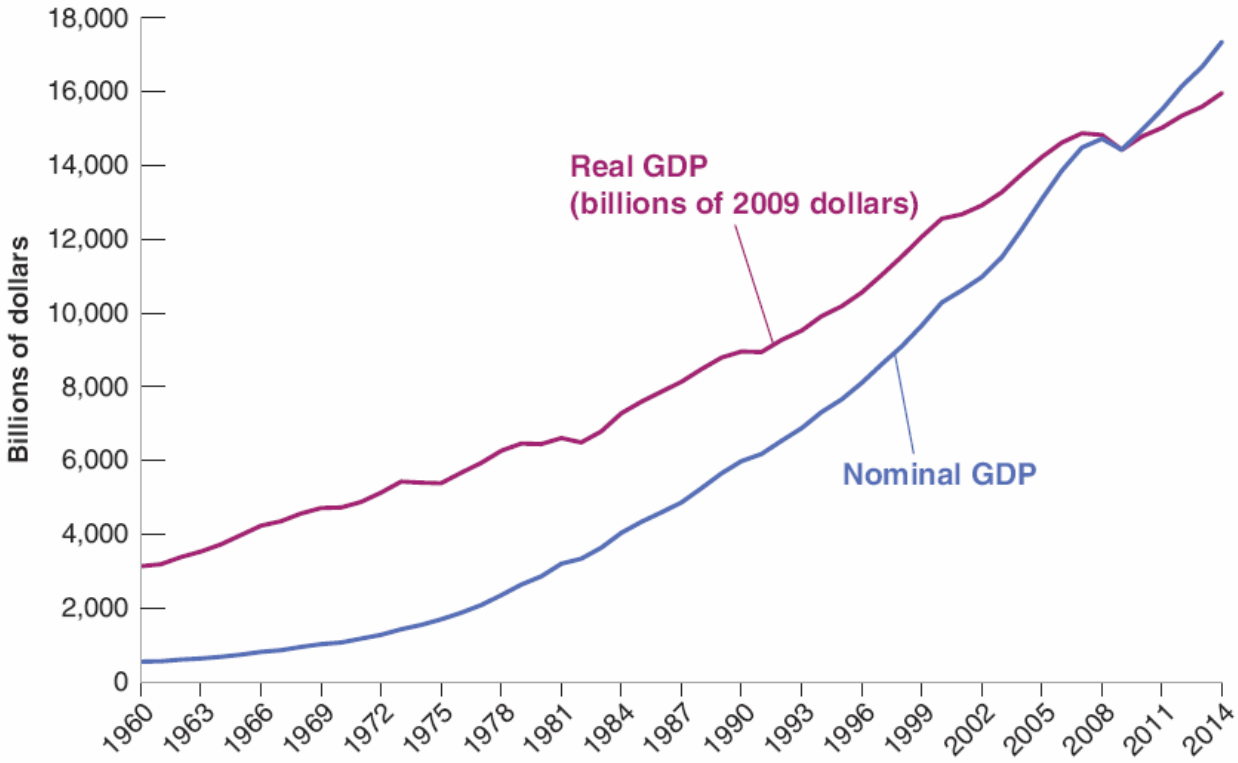
- **Real GDP** is the sum of quantities of final goods times *constant* (not *current*) prices. Is also called **GDP in terms of goods, GDP in constant dollars, GDP adjusted for inflation,** or **GDP in chained (2009) dollars**, or **GDP in 2009 dollars**.

https://www.youtube.com/watch?v=rGqhTQyY6g4

1. Aggregate Output

https://fred.stlouisfed.org/graph/?g=gWU

Figure 2-1 Nominal and Real U.S. GDP, 1960–2014



Source: Series GDPCA, GDPA: Federal Reserve Economic Data (FRED) http://research.stlouisfed.org/fred2/.

From 1960 and 2014,

Nominal GDP increased by a factor of 32.

Real GDP increased by a factor of about 5.

1. Aggregate Output

EXAMPLE 3

Calculate nominal GDP for 2012 and 2013.

Calculate real GDP for 2012 and 2013.

	Peanut Butter		Jelly	
Year	**Quantity**	**Price**	**Quantity**	**Price**
2012	20	$4.00	50	$2.00
2013	30	$5.00	100	$2.00

1. Aggregate Output

Nominal GDP for 2012:

$$P_{pb}^{12} \times Q_{pb}^{12} + P_{jelly}^{12} \times Q_{jelly}^{12} = \$4.00 \times 20 + \$2.00 \times 50$$
$$= \$80 + \$100 = \$180$$

Nominal GDP for 2013:

$$P_{pb}^{13} \times Q_{pb}^{13} + P_{jelly}^{13} \times Q_{jelly}^{13} = \$5.00 \times 30 + \$2.00 \times 100$$
$$= \$150 + \$200 = \$350$$

1. Aggregate Output

Real GDP for 2012:

$$P_{pb}^{12} \times Q_{pb}^{12} + P_{jelly}^{12} \times Q_{jelly}^{12} = \$4.00 \times 20 + \$2.00 \times 50$$
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Real GDP for 2013:

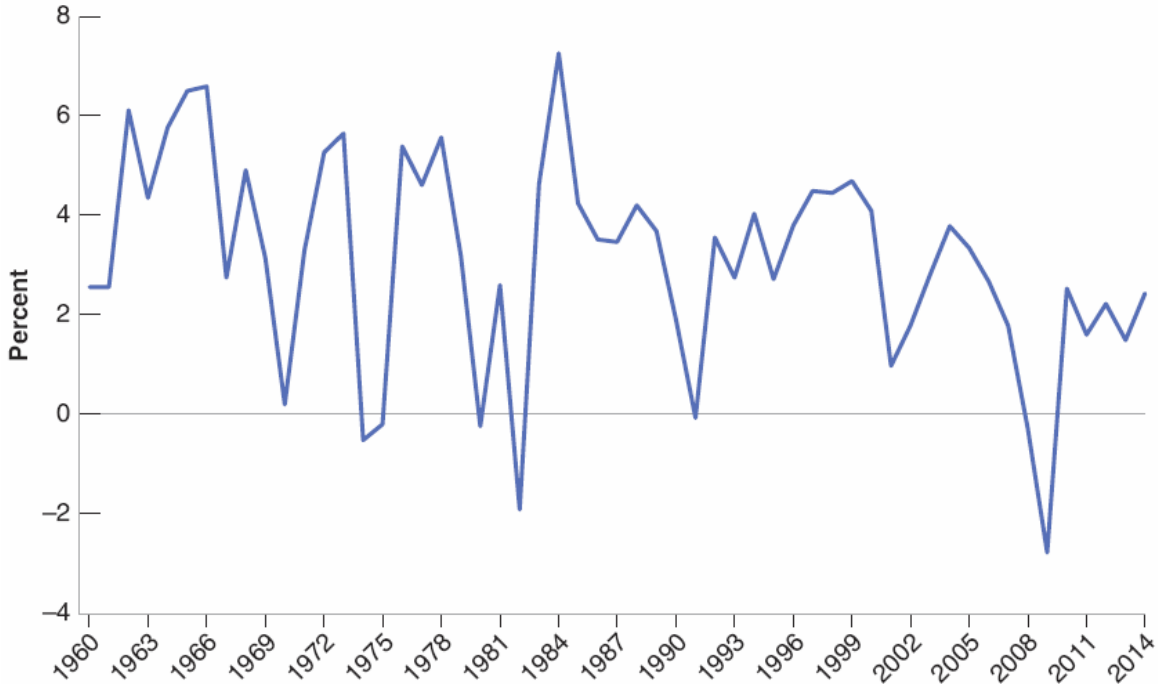
$$P_{pb}^{12} \times Q_{pb}^{13} + P_{jelly}^{12} \times Q_{jelly}^{13} = \$4.00 \times 30 + \$2.00 \times 100$$
$$= \$120 + \$200 = \$320$$

1. Aggregate Output

GDP growth in year t is $(Y_t - Y_{t-1})/Y_{t-1}$.

Since 1960, the U.S. economy has gone through a series of expansions, interrupted by short recessions.

Figure Growth Rate of U.S. GDP, 1960–2014

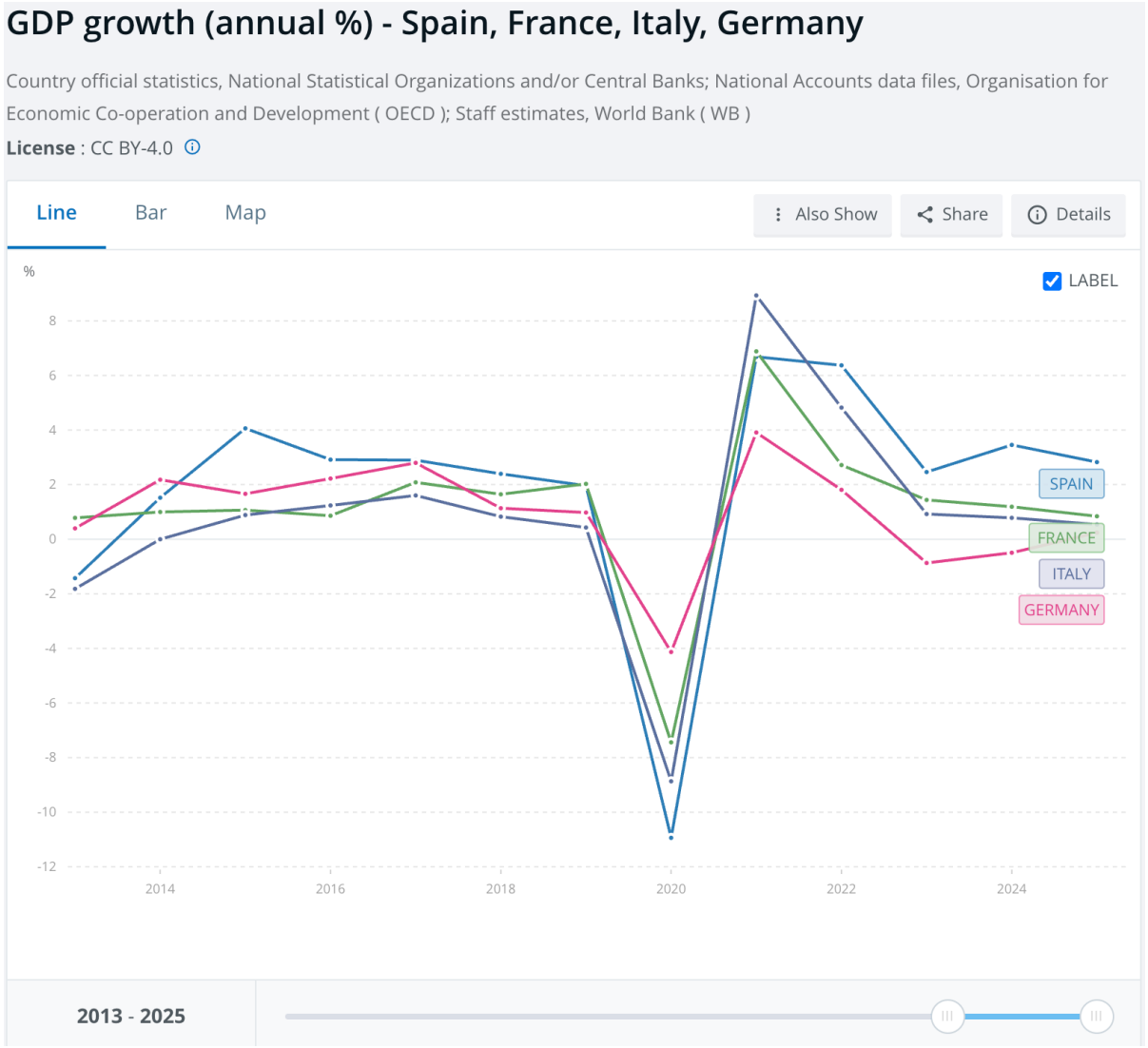


Source: Calculated using series GDPCA in Figure 2-1.



WHICH COUNTRY GREW THE MOST?

Source: World bank https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG?end=2025&locations=ES-FR-IT-DE&start=2013



WHICH COUNTRY GREW THE MOST?

https://data.worldbank.org/indicator/NY.GDP.PCAP.KD.ZG?end=2024&locations=ES-DE-FR-IT&start=2018

What is the Growth Rate of the Spanish GDP between 2018 and 2024?

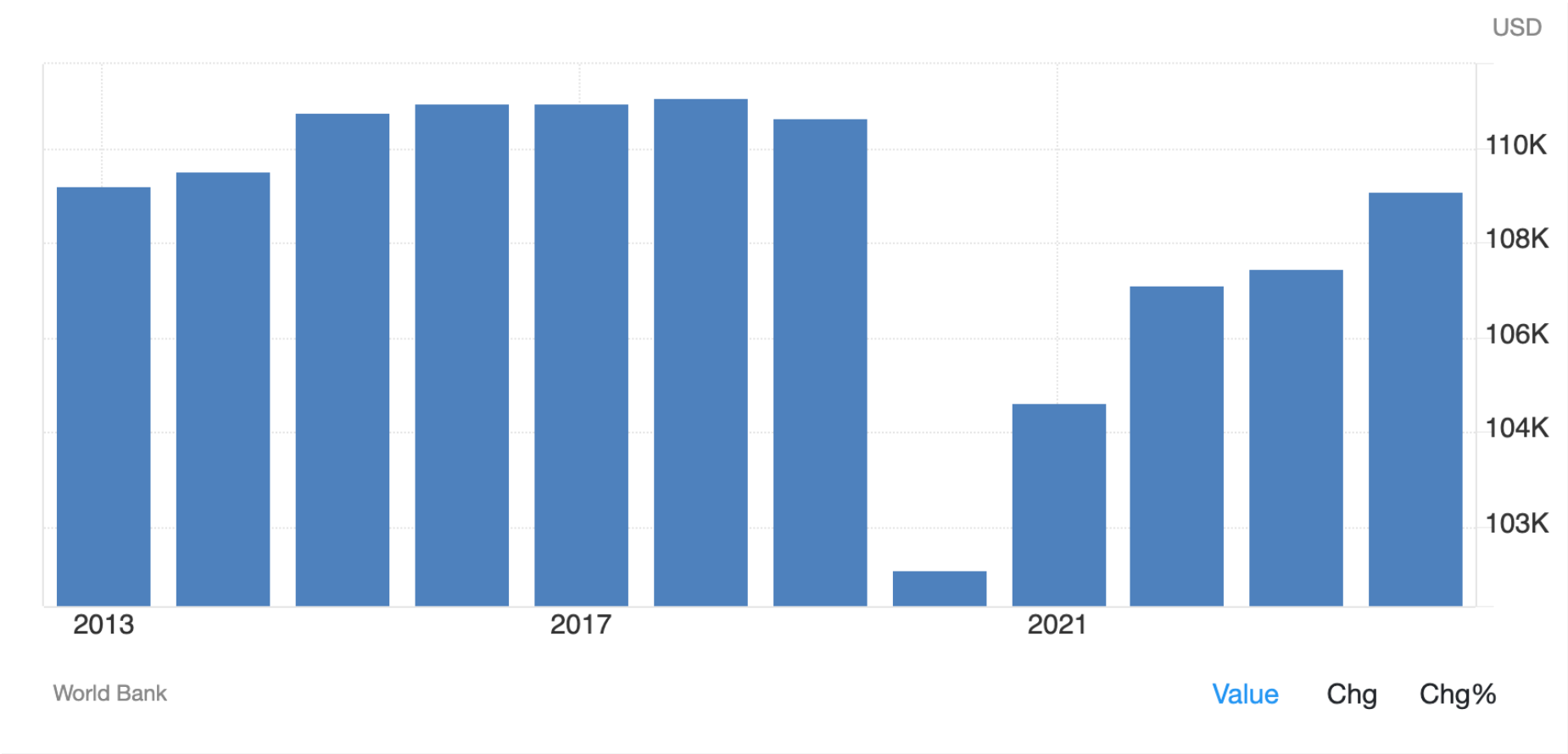
Compound growth rate

$$\text{CAGR} = \left(\frac{\text{Ending Value}}{\text{Beginning Value}} \right)^{\frac{1}{t}} - 1$$


IS SPAIN A GOOD COUNTRY TO LIVE?

GDP per capita (PPP) for Spain





Spain - GDP Per Person Employed (constant 1990 PPP $)

Question: Nominal and real GDP

Eden, a country, produces just two output goods: apples and snakeskin handbags. In 2014 the prices of apples and handbags were $10 and $24, respectively. In 2015, the prices increased to $12 and $25. The quantities produced of the two goods were 100 and 20, respectively, in both 2014 and 2015. From this information, we can conclude that:

Select all correct answers

a. The nominal GDP in 2014 is $1,480.
b. The real GDP is unchanged between 2014 and 2015.
c. The nominal GDP growth rate between 2014 and 2015 is 14.9%.
d. The economy of Eden grew between 2014 and 2015 in real terms.

ANSWER: Nominal and real GDP

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Feedback

a. The nominal GDP in 2014 is (10 x 100) + (24 x 20) =$1480.
b. The quantities produced (total output) did not change between 2014 and 2015, so real GDP also did not change.
c. The nominal GDP in 2014 is $1480 and the nominal GDP in 2015 is $1700. The growth rate is therefore (1700 – 1480)/1480 = 14.9%.
d. Here, the increase in the nominal GDP between the two years is purely due to the price increase. Therefore in real terms the economy did not grow.

Active Learning: *GDP* and its components

What is the effect on ***GDP*** and its components?

A. Jahzara spends $300 to buy her husband dinner at the finest restaurant in Boston.

B. Nylah spends $1,200 on a new smartphone to use in her publishing business. The smartphone was built in China.

C. Joseph spends $800 on a tablet to use in his editing business. He got last year's model on sale for a great price from a local manufacturer.

D. GM builds $500 million worth of cars, but consumers only buy $470 million worth of them.

A. Jahzara spends $300 to buy her husband dinner at the finest restaurant in Boston.
- ***C*** rises by $300
- ***GDP*** rises by $300.

B. Nylah spends $1,200 on a new smartphone to use in her publishing business. The smartphone was built in China.
- ***I*** rises by $1,200
- ***NX*** fall by $1,200 (because Imports rise by $1,200)
- ***GDP*** is unchanged.

C. Joseph spends $800 on a tablet to use in his editing business. He got last year's model on sale for a great price from a local manufacturer.
- ***I*** and ***GDP*** are unchanged
  - ***I*** (Joseph's business) rises by $800
  - ***I*** (inventories) fall by $800

D. GM builds $500 million worth of cars, but consumers only buy $470 million of them.
- ***C*** rises by $470 million
- ***I*** (inventory) rises by $30 million
- ***GDP*** rises by $500 million.

2. The Unemployment Rate

- **Employment** is the number of people who have a job.
- **Unemployment** is the number of people who do not have a job but are looking for one.
- The **labor force** is the sum of employment and unemployment.

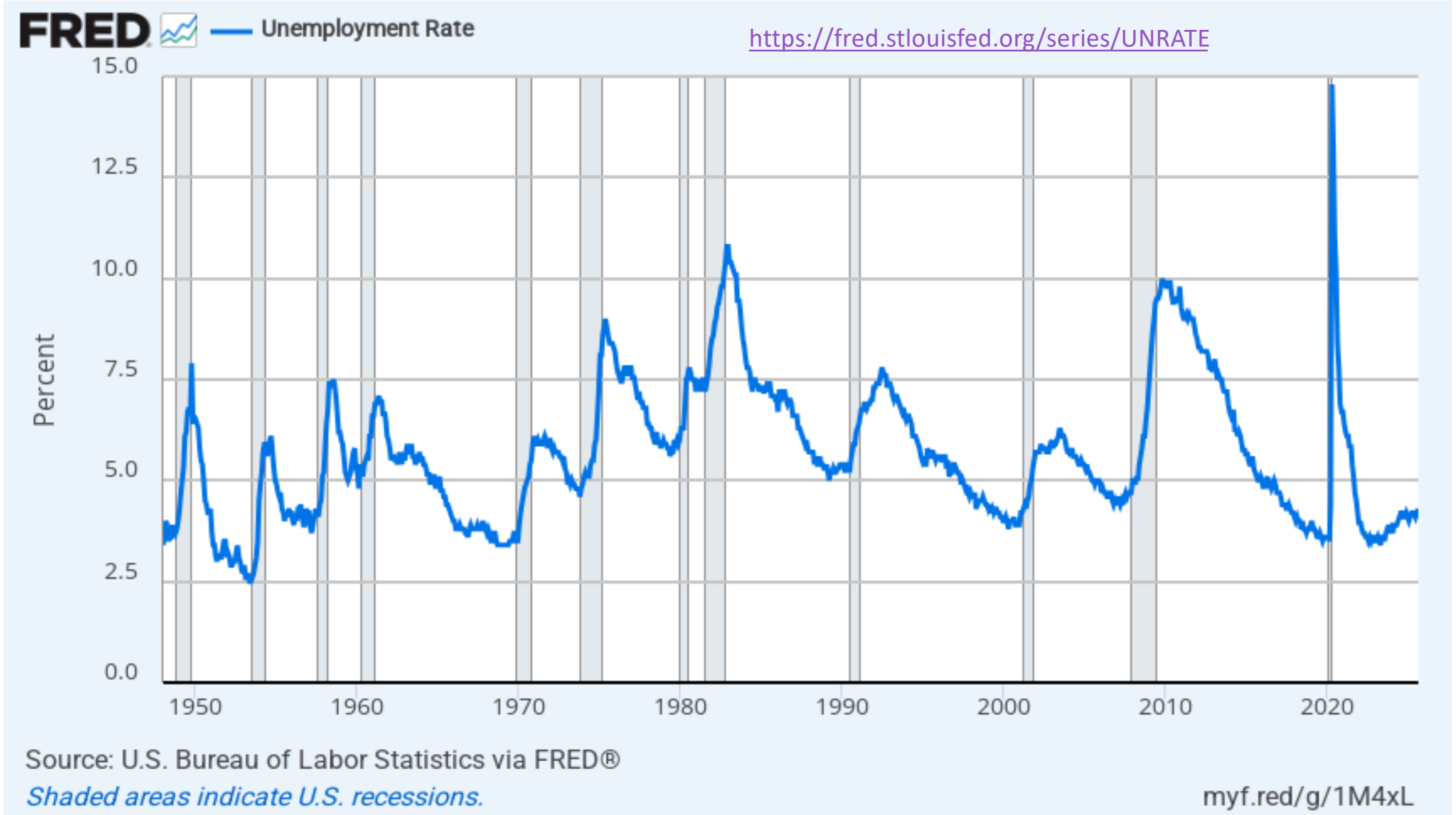
$$L = N + U$$

$$\text{labor force} = \text{employment} + \text{unemployment}$$

- The **unemployment rate** is the ratio of the number of people who are unemployed to the number of people in the labor force.

$$u = \frac{U}{L}$$

$$\text{unemployment rate} = \text{unemployment} / \text{labor force}$$


FRED
Unemployment Rate
https://fred.stlouisfed.org/series/UNRATE
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Percent
1950
1960
1970
1980
1990
2000
2010
2020
Source: U.S. Bureau of Labor Statistics via FRED®
Shaded areas indicate U.S. recessions.
myf.red/g/1M4xL

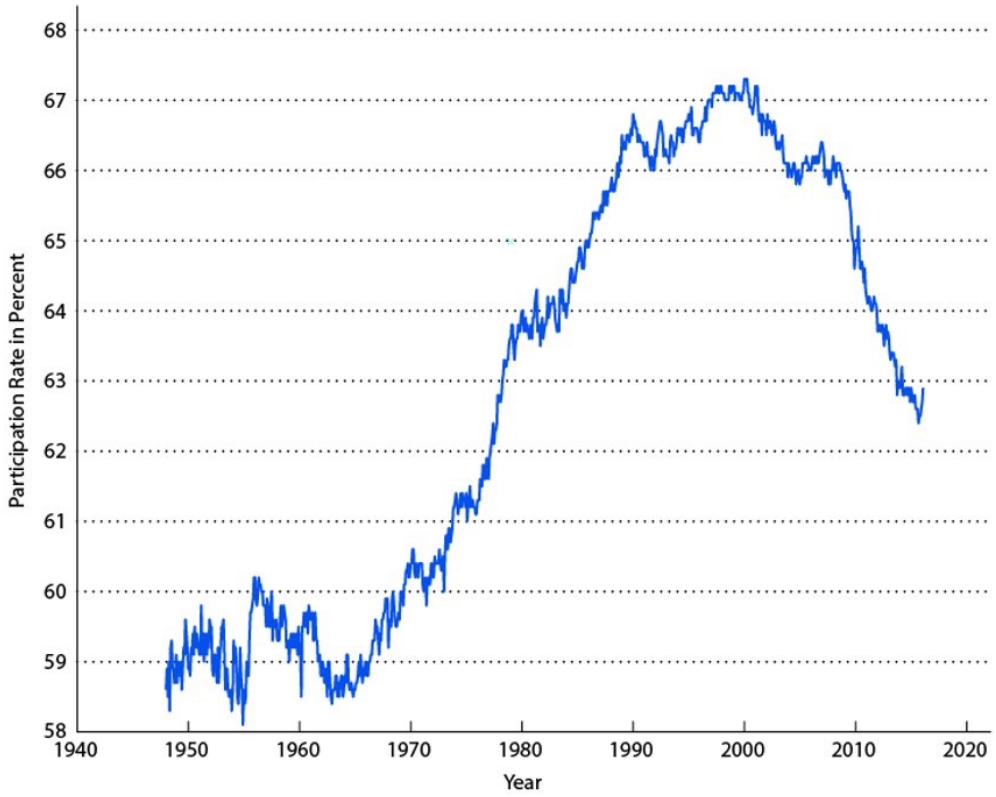
2. The Unemployment Rate

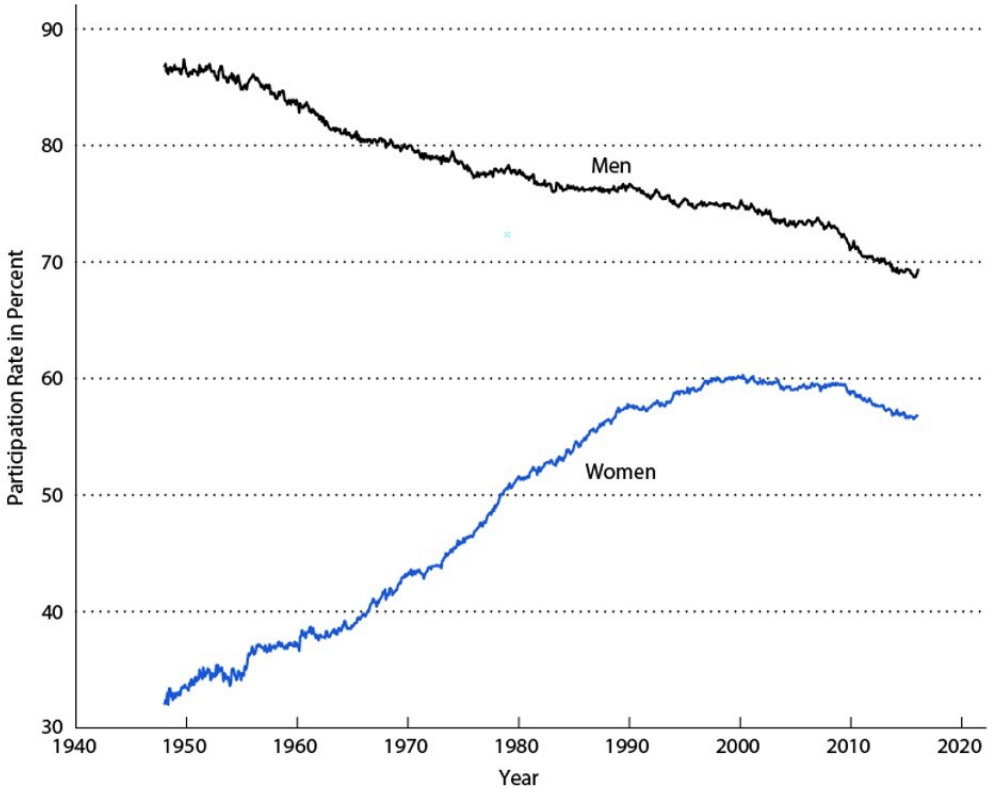
Discouraged workers are those who give up looking for a job and so no longer counted as unemployed.

The **Labor Force Participation rate** is the ratio of the labor force to the total population of working age (15-65 years old).

Because of discouraged workers, a higher unemployment rate is typically associated with a lower participation rate.

Labor Force Participation Rates in the US





Source: Williamson (2018)

Exercise: Unemployment and Labor Force Participation in Spain

Suppose the following data describe Spain's working-age population in a year:

- Employed people: 21 million
- Unemployed people who are actively seeking work: 2 million
- People who are not working and are not seeking work: 17 million

Answer the following questions:

1. Calculate Spain's labor force.
2. Calculate the unemployment rate.
3. Calculate the labor force participation rate.
4. Explain why people who are not actively seeking work are not counted as unemployed.

Answer Key

1. Labor force
Labor force = 21 million employed + 2 million unemployed = **23 million people**

2. Unemployment rate
Unemployment rate = (2 million ÷ 23 million) × 100 = **8.7%**

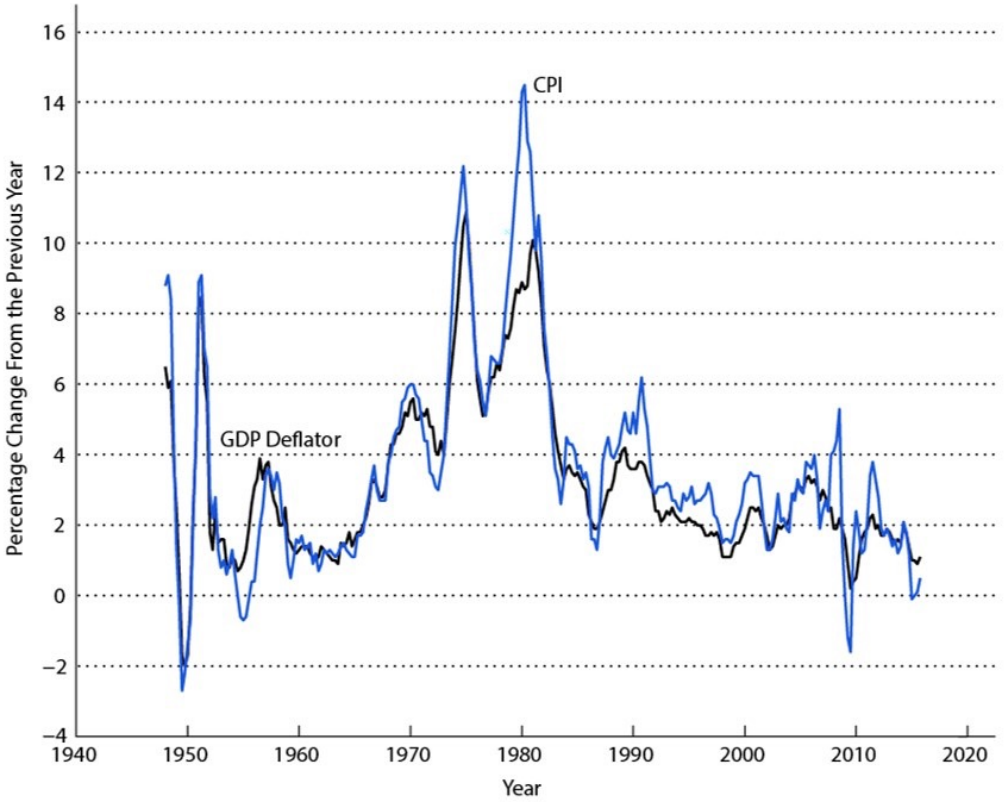
3. Labor force participation rate
Working-age population = 21 million + 2 million + 17 million = **40 million people**
Labor force participation rate = (23 million ÷ 40 million) × 100 = **57.5%**

4. Explanation
People who are not working and are not actively looking for a job are classified as **outside the labor force**, not as unemployed. To be considered unemployed, a person must be without work, available to work, and actively seeking employment.

3. The Inflation Rate

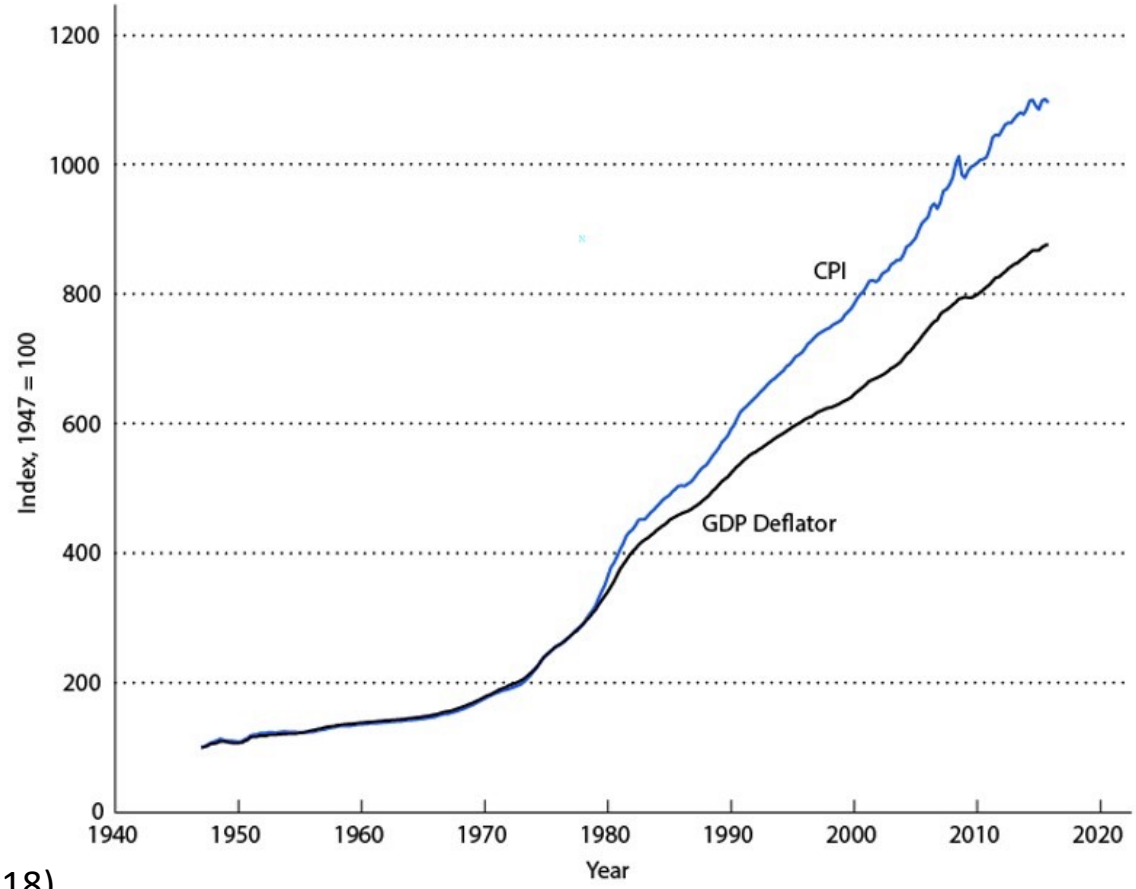
- **Inflation** is a sustained rise in the general level of prices—the **price level**.
- The **inflation rate** is the rate at which the price level increases.
- **Deflation** is a sustained decline in the price level (negative inflation rate).
- **Disinflation** is a reduction in the inflation rate.

Inflation Rate Calculated from the CPI & from the Implicit GDP Price Deflator



Source: Williamson (2018)

The Price Level as Measured by the CPI and the Implicit GDP Price Deflator



Source: Williamson (2018)

3. The Inflation Rate

The **GDP deflator** in year *t* (P_t) is the ratio of nominal GDP to real GDP in year *t*:

$$P_t = \frac{\text{Nominal GDP}_t}{\text{Real GDP}_t} = \frac{\$Y_t}{Y_t}$$

It is called an **index number** (1 in 2009), which has **no economic interpretation.**

The rate of change has a clear interpretation: the rate of inflation.

$$\pi_t = (P_t - P_{t\text{-}1})/P_{t\text{-}1}$$

EXERCISE

If the price index is 200 in Year 2012 and 210 in Year 2013, the rate of inflation is:

a. 20%.
b. 5%.
c. 4.76%.

$$\text{Inflation rate in 2013} = \frac{(\text{Price index in 2013}) - (\text{Price index in 2012})}{(\text{Price index in 2012})} \times 100$$

Answer: ***b*** *– the rate of inflation is (210-200)/200 x 100 = 5%.*

3. The Inflation Rate

- The set of goods produced in the economy is not the same as the set of goods purchased by consumers because:
  - Some of the goods in GDP are sold not to consumers but to firms, to the government, or to foreigners.
  - Some of the goods bought by consumers are not produced domestically but are imported from abroad.

3. The Inflation Rate

- The **Consumer Price Index** (**CPI**) is a measure of the **cost of living**.

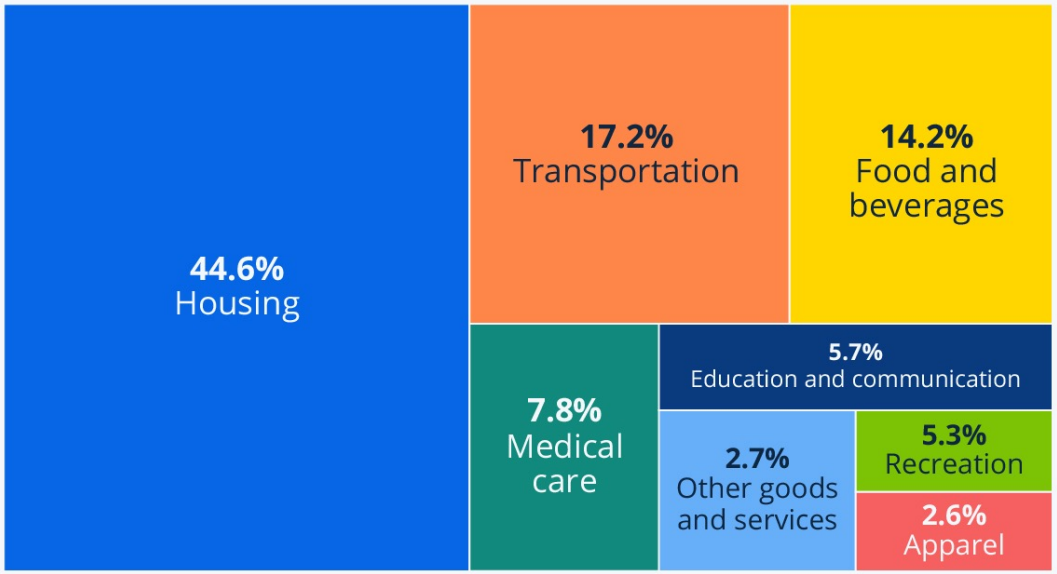
- In the US: The CPI is published monthly by the Bureau of Labor Statistics (BLS), which collects price data for a representative basket of roughly 80,000 goods and services.

- In Spain: Around 487 traditional products and 263 scanner-data items are tracked monthly via establishment interviews, automated scanner data, and centralized collection.

- The CPI gives the cost in dollars of a specific list of goods and services over time.

The Components of the Consumer Price Index

Relative importance of expenditure categories in the Consumer Price Index for All Urban Consumers



As of September 2023
Source: U.S. Bureau of Labor Statistics

statista

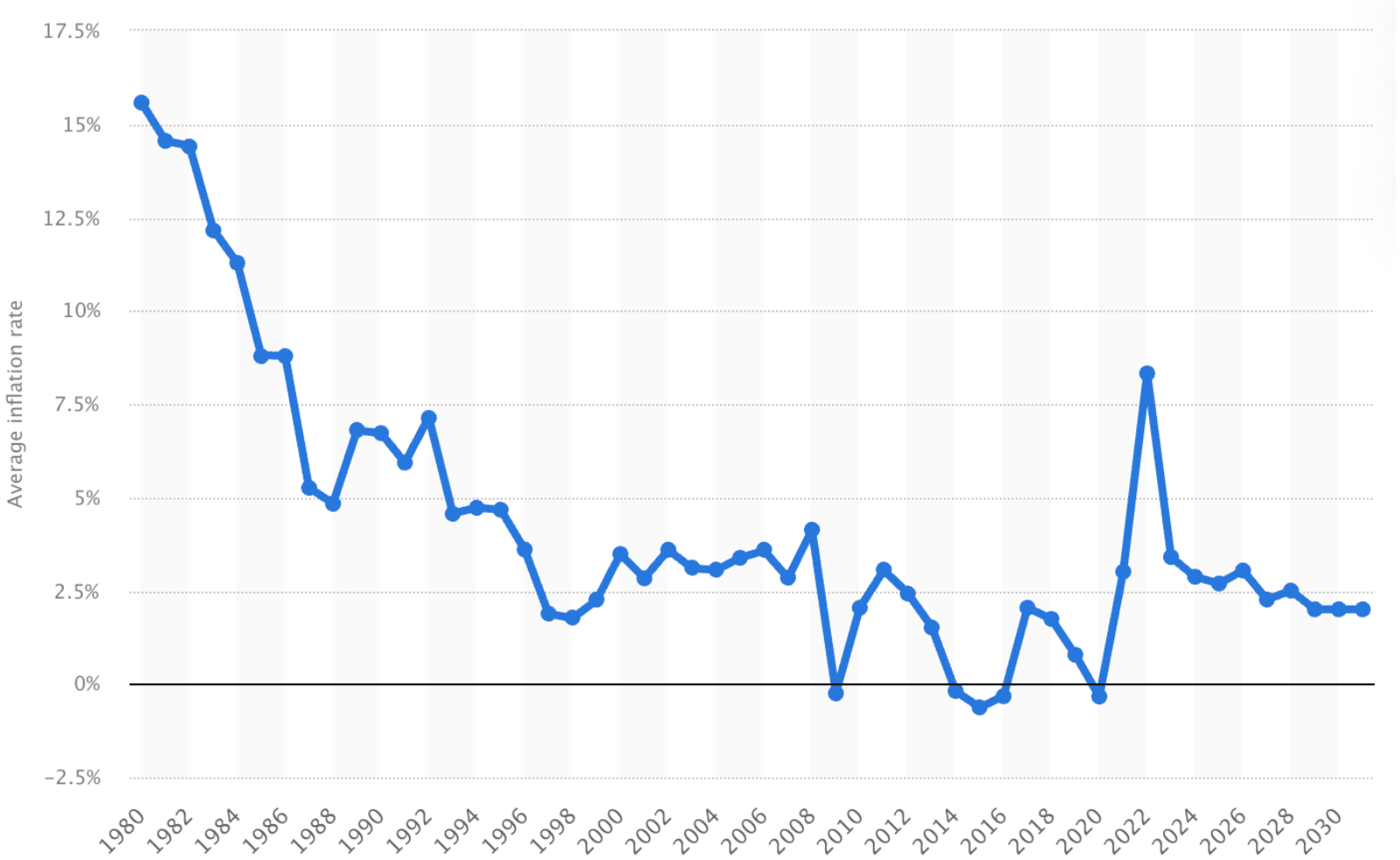
3. The Inflation Rate

The GDP deflator and the CPI formula look nearly identical.

Question: What are the differences?

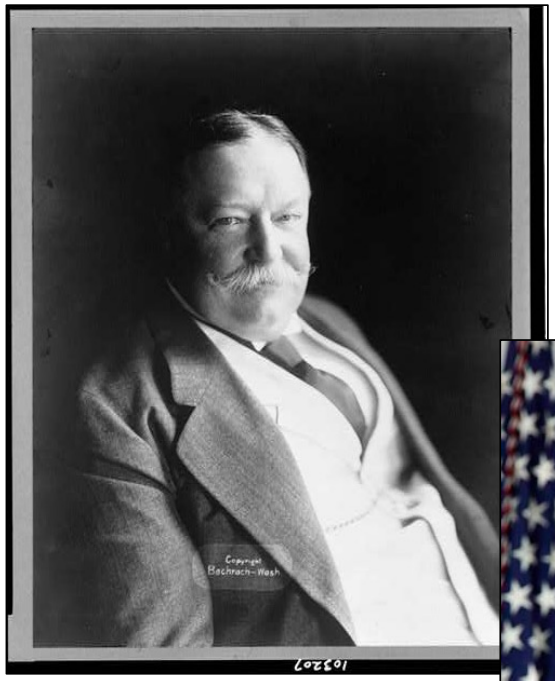
1. The GDP deflator includes things not purchased by households, like trains, subways, and submarines.
2. The CPI includes imports like Chinese laptops.
3. Housing-related expenditures like shelter and utility bills have a large weight in the CPI.

Inflation rate in Spain (CPI)



Source: Statista

3. The Inflation Rate





We can use a price index to make meaningful comparisons across time:

- In 1909, then U.S. President William Howard Taft was paid **$75,000.**

- In 2013, current U.S. President Barack Obama was paid **$400,000.**

3. The Inflation Rate

We can convert Taft's salary to 2013 dollars by applying this formula:

$$\text{Value in 2013 dollars} = \frac{\text{(Price index in 2013)}}{\text{(Price index in 1909)}} \times \text{Value in 1909 dollars}$$

$$= \frac{233}{9} \times \$75{,}000$$

$$= \$1.9 \text{ million}$$

ECON PRACTICE



Calculating Inflation

PLAY

ECON PRACTICE



Winners and Losers of Inflation

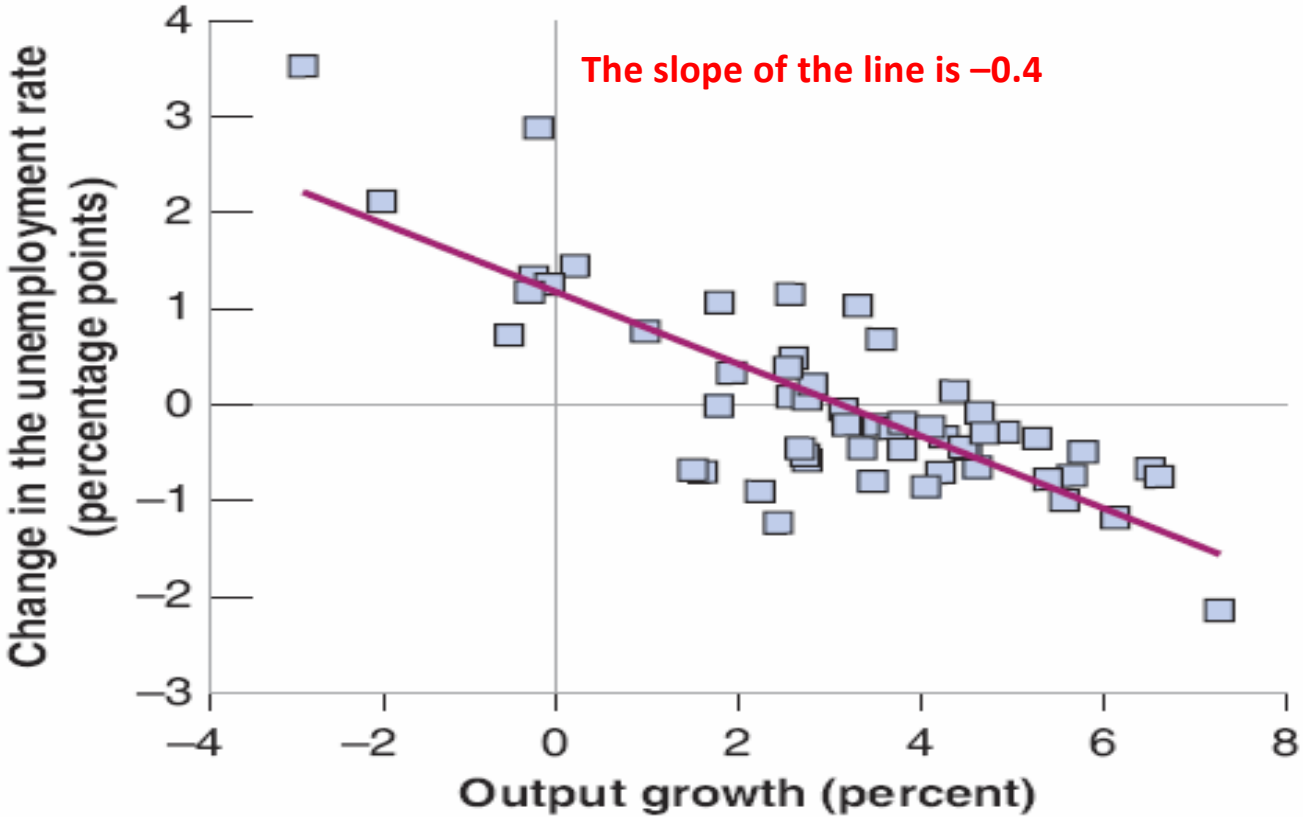
PLAY

4. Okun's Law: Output and Unemployment

Figure 2-5 Changes in the Unemployment Rate versus Growth in the US, 1960–2014

The line crosses the horizontal axis where output growth is 3%, meaning that

it takes a growth rate of 3% to **keep unemployment constant**.



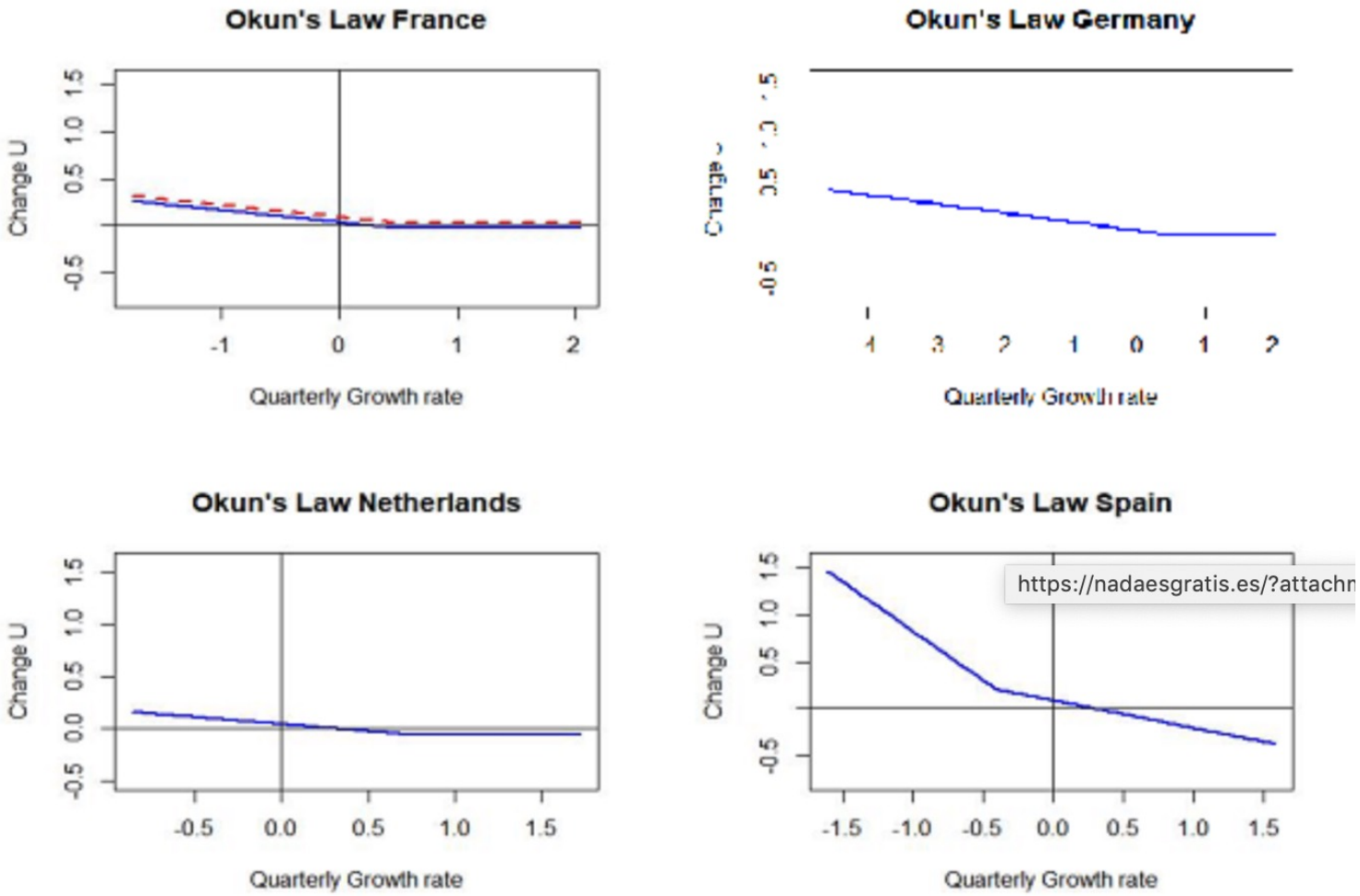


Figure 5. Comparative of Okun's law across countries (Spain scale).

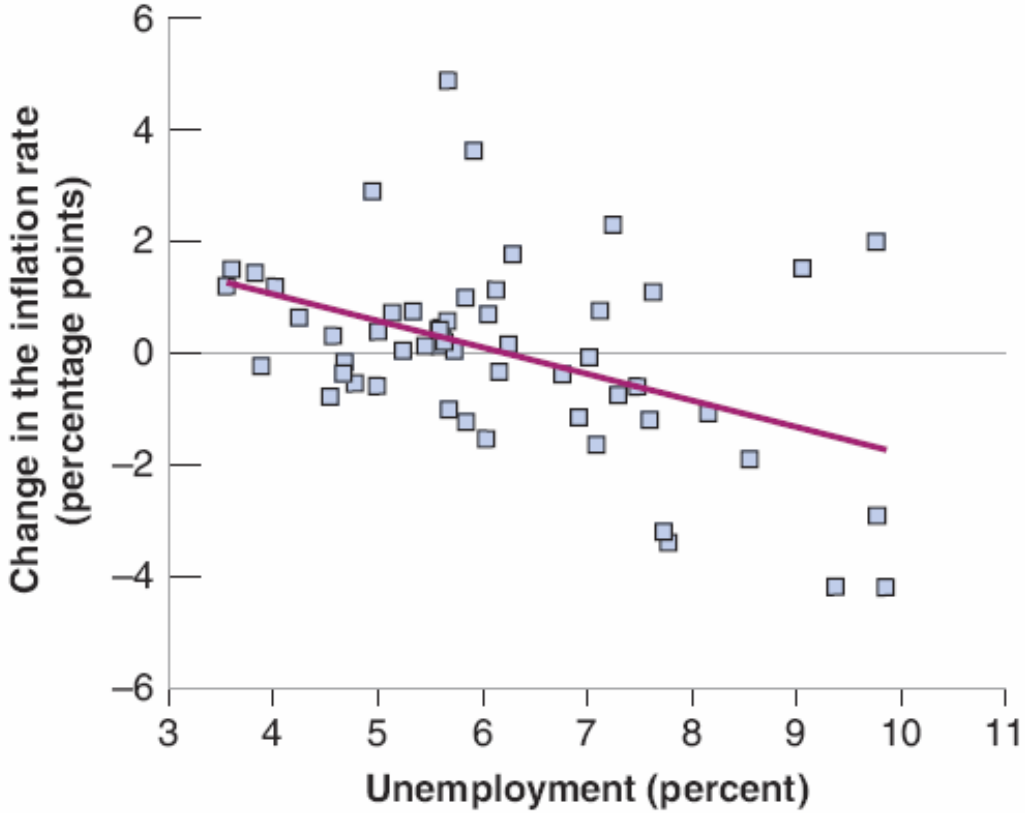
Source: C. Nebot, A. Beyaert y J. García-Solanes (2019), "New insights into the nonlinearity of Okun's law"

5. Phillips Curve: Unemployment and the Inflation Rate

Figure 2-6 Changes in the Inflation Rate versus the Unemployment Rate in the US, 1960–2014

The line crosses the horizontal axis where the unemployment rate is equal to about 6%,

meaning that inflation typically increased when unemployment was below 6%.



5. Phillips Curve: Unemployment and the Inflation Rate

Figure 1 The Phillips Curve, 1961–1969

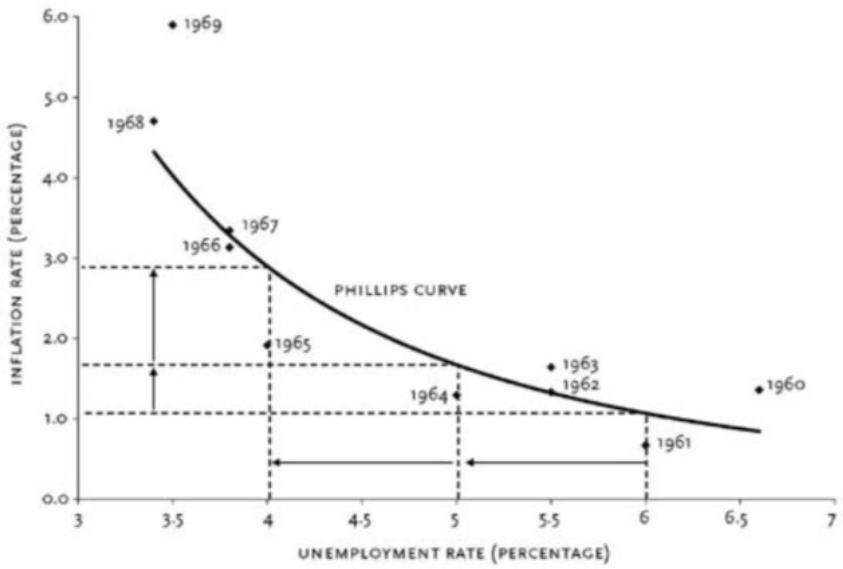
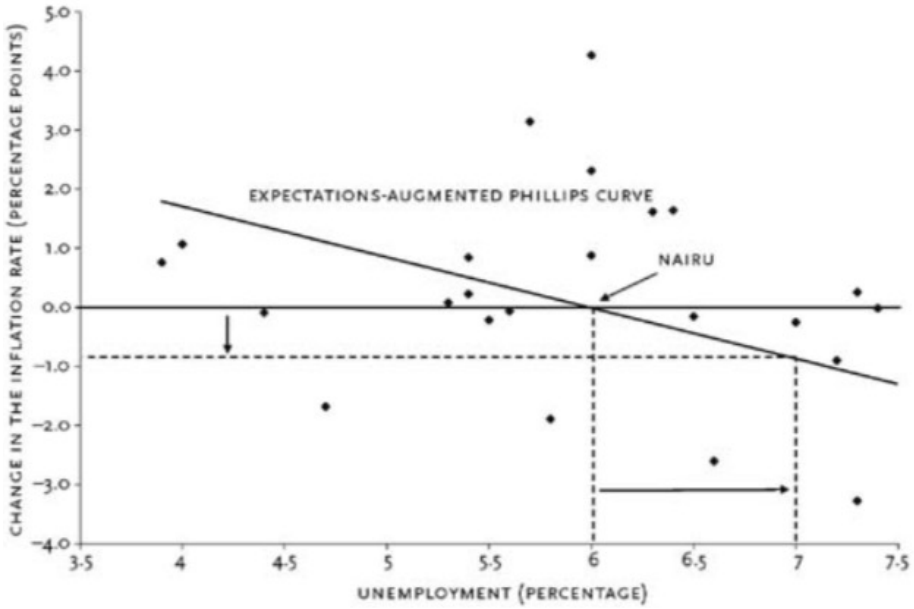


Figure 2 The Expectations-Augmented Phillips Curve, 1976–2002



https://www.econlib.org/library/Enc/PhillipsCurve.html

Videos about Key Macro Concepts

https://mru.org/principles-economics-macroeconomics-0

THAT IS ALL!